

SUPPLEMENT TO “IDENTIFYING RATIONAL TYPES IN UNKNOWN
ENVIRONMENTS”
APPENDIX 3: FORMAL PROOFS AND LEAN VERIFICATION

HUMBERTO BERNAL

This supplement presents in full detail the formal statements and proofs supporting the identification argument in the main paper. It is organized in eight sections that mirror the logical structure of the model: stochastic trends (Section ??), public information and Bayesian learning (Section ??), the stochastic implementation mechanism (Section ??), asymptotic type identification (Section ??), competitive risks and operational pressure (Section ??), policy implications (Section ??), equilibrium implementability (Section ??), and the abstract Kakutani identification criterion (Section ??), plus machine-checked verification (Section ??). Analytical proofs are self-contained in this document; algebraic steps verified in Lean 4 are reproduced from the `Appendix_3/` replication package (Section ??). Classical measure-theoretic inputs (Kakutani, Blackwell–Dubins, fixed-point existence) remain proved here and are not re-derived inside the proof assistant.

S1. RESULTS FROM THE MAIN PAPER

The following result is imported from *Identifying Rational Types in Unknown Environments: Evidence from Kidnapping in Colombia*. It operates in the time-series domain and is independent of the game-theoretic blocks below. Lemma ?? supplies the integration-order step.

LEMMA S1.1— $I(0)$ closed under linear combinations: *If $X_t \sim I(0)$ and $Y_t \sim I(0)$, then $aX_t + bY_t \sim I(0)$ for all $a, b \in \mathbb{R}$.*

PROOF: Covariance stationarity is preserved under finite linear combinations: for all s, t and all $a, b \in \mathbb{R}$,

$$\mathbb{E}[(aX_{t+s} + bY_{t+s}) - (aX_t + bY_t)] = 0,$$

and the autocovariance function of $aX_t + bY_t$ depends on lags only through $t - s$. Hence $aX_t + bY_t$ is integrated of order zero. *Q.E.D.*

COROLLARY S1.2—Equivalence of stochastic trends: *Let $\{y_t\}$ be a time series integrated of order one, $I(1)$. Suppose the existence of two valid decompositions:*

$$y_t = P_t^{(1)} + u_t^{(1)} = P_t^{(2)} + u_t^{(2)},$$

where $P_t^{(1)}$ and $P_t^{(2)}$ are $I(1)$ processes and $u_t^{(1)}, u_t^{(2)}$ are stationary $I(0)$ processes. Then:

$$P_t^{(1)} - P_t^{(2)} \sim I(0).$$

Economist, professor of the Economics Program at the Universidad Colegio Mayor de Cundinamarca and graduate of the Ph.D. in Economics from the Universidad de los Andes, Colombia.

Consequently, all valid stochastic trends derived from the same observed series are equivalent up to a stationary component.

PROOF: From the two decompositions, $P_t^{(1)} - P_t^{(2)} = u_t^{(2)} - u_t^{(1)}$. Since $u_t^{(1)}$ and $u_t^{(2)}$ are $I(0)$, Lemma ?? with $a = 1$ and $b = -1$ yields $P_t^{(1)} - P_t^{(2)} \sim I(0)$. Any pair of valid stochastic trends extracted from the same $I(1)$ series therefore differs by a transitory $I(0)$ component. *Q.E.D.*

S2. PUBLIC INFORMATION, LIKELIHOOD, AND BAYESIAN LEARNING

The three definitions below construct the public filtration, the likelihood kernel, and the belief process. The stability lemma closes the block.

DEFINITION S2.1—Public signal and filtration: **(i) Public signal** ζ_t . The public signal ζ_t is the random vector that summarizes what is observable at the close of period t :

$$\zeta_t = (\alpha_t, \gamma_t, \tilde{a}_t^F, \tilde{a}_t^K, \tilde{a}_t^S, m_t, d_t) \in \mathcal{Z}_t \subseteq \mathbb{R}^2 \times \mathcal{A}^F \times \mathcal{A}^K \times \mathcal{A}^{S, \text{disc}} \times \mathcal{M}_{\text{out}} \times \{0, 1\}.$$

(ii) Public filtration $\{\mathcal{F}_t^{\text{pub}}\}$. Set $\mathcal{F}_0^{\text{pub}} := \sigma(z, \theta_V)$. For $t \geq 1$,

$$\mathcal{F}_t^{\text{pub}} := \sigma(z, \zeta_0, \dots, \zeta_{t-1}),$$

that is, the σ -algebra generated by the initial location and by the signals observed strictly before period t begins.

(iii) Measurability. Each $\zeta_t : (\Omega, \mathcal{F}) \rightarrow (\mathcal{Z}_t, \mathcal{B}(\mathcal{Z}_t))$ is measurable, and $\sigma(z, \zeta_0, \dots, \zeta_{t-1})$ is well defined as a sub- σ -algebra of $\mathcal{F}$. □

DEFINITION S2.2—Predictive likelihood under a counterfactual type: Fixing a PBE with profile (χ, s^F, s^K) and a counterfactual type $\theta \in \Theta_K$, the **predictive likelihood** of period t is

$$\ell_t(\theta; J) := \mathbb{P}_{C, \theta}(\zeta_t \in J \mid \mathcal{F}_t^{\text{pub}}), \quad J \in \mathcal{B}(\mathcal{Z}_t). \quad (\text{S2.1})$$

The mapping $J \mapsto \ell_t(\theta; J)$ is a **probability kernel** that carries the information state $\mathcal{F}_t^{\text{pub}}$ to the law of ζ_t under type θ . □

DEFINITION S2.3—Beliefs and PBE-consistent Bayesian updating: Let $\mu_0 \in \Delta(\Theta_K)$ be the **common prior** with full support, $\mu_0(\theta \mid z, \theta_V) > 0$ for all $\theta \in \Theta_K$. For $t \geq 1$, let $\mu_t \in \Delta(\Theta_K)$ be the distribution obtained by iterated Bayes from μ_0 , using the realizations of $(\zeta_0, \dots, \zeta_{t-1})$ and the predictive kernels $\ell_s(\theta'; J)$ induced by the same PBE.

The process $\{\mu_t\}_{t \geq 0}$ is **consistent with the PBE** if, under the ex-ante law $\mathbb{P}_E$,

$$\mu_t(\theta) = \mathbb{P}_E(\theta_K = \theta \mid \mathcal{F}_t^{\text{pub}}) \quad \forall \theta \in \Theta_K, \forall t \geq 1, \quad (\text{S2.2})$$

interpreted $\mathbb{P}_E$ -almost surely upon fixing regular versions of the conditional probabilities. □

LEMMA S2.4—Asymptotic stability of beliefs. Convergence theorem of **?, ?**: *Under the ex-ante law $\mathbb{P}_E$ of Definition **??**, the process $\{\mu_t(\theta)\}_{t \geq 0}$ is a **bounded martingale** that converges $\mathbb{P}_E$ -almost surely to a limit random variable $Z_\theta \in [0, 1]$ $\mathbb{P}_E$ -a.s.*

PROOF: The proof verifies the conditions of the convergence theorem of **???**.

1. Martingale property. Let $M = \mathbf{1}\{\theta_K = \theta\}$. By equilibrium consistency (Definition **??**),

$$\mu_t(\theta) = \mathbb{E}^{\mathbb{P}_E}[M \mid \mathcal{F}_t^{\text{pub}}] \quad \mathbb{P}_E\text{-a.s.}$$

As $\mathcal{F}_t^{\text{pub}} \subseteq \mathcal{F}_{t+1}^{\text{pub}}$, the **law of iterated expectations** yields

$$\mathbb{E}^{\mathbb{P}_E}[\mu_{t+1}(\theta) \mid \mathcal{F}_t^{\text{pub}}] = \mathbb{E}^{\mathbb{P}_E}[\mathbb{E}^{\mathbb{P}_E}[M \mid \mathcal{F}_{t+1}^{\text{pub}}] \mid \mathcal{F}_t^{\text{pub}}] = \mathbb{E}^{\mathbb{P}_E}[M \mid \mathcal{F}_t^{\text{pub}}] = \mu_t(\theta) \quad \mathbb{P}_E\text{-a.s.}$$

The process is a martingale under $\mathbb{P}_E$.

2. Pointwise and uniform L^1 bound. As $\mu_t(\theta)$ is a probability mass, $0 \leq \mu_t(\theta) \leq 1$ for all $t \geq 0$ (outside a $\mathbb{P}_E$ -null set). In particular, $\sup_{t \geq 0} \mathbb{E}^{\mathbb{P}_E}[|\mu_t(\theta)|] \leq 1$, which provides the uniform L^1 bound required by Doob's theorem.

3. Almost sure convergence. Once the martingale property and the uniform L^1 bound are verified, **Doob's convergence theorem** guarantees the existence of Z_θ such that

$$\lim_{t \rightarrow \infty} \mu_t(\theta) = Z_\theta \quad \mathbb{P}_E\text{-a.s.}$$

The pointwise bound $0 \leq \mu_t(\theta) \leq 1$ implies $Z_\theta \in [0, 1]$ $\mathbb{P}_E$ -a.s.

Q.E.D.

S3. STOCHASTIC IMPLEMENTATION MECHANISM (MDG)

This section establishes the fundamental definitions of the stochastic implementation layer (MDG), which transforms strategic intentions into materialized actions, and of the law governing the physical outcome that determines the observable results of the episode. The episode horizon is the endogenous stopping time

$$\tau := \inf\{t \geq 0 : m_t \in \mathcal{M}_{\text{term}}\}, \quad (\text{S3.1})$$

where m_t is the physical outcome, $\mathcal{M}_{\text{cont}} = \{\text{cont}\}$, $\mathcal{M}_{\text{term}} = \{\text{pay, res, rel, kill}\}$, and $\mathcal{M}_{\text{out}} = \mathcal{M}_{\text{cont}} \cup \mathcal{M}_{\text{term}}$ (fully formalized in Definition **??**). Active periods satisfy $t < \tau$ while captivity continues.

DEFINITION S3.1—MDG stochastic implementation filter: Let $i \in \{K, S, F\}$ and let $\mathcal{A}^i$ be the finite set of admissible actions of agent i . The **state vector** is $X_t := (\mu_t, h_t)$, with μ_t from Definition **??** and h_t the public history at the opening of t . Given the plan $a_t^{i*} \in \mathcal{A}^i$ and X_t , execution is determined by the **implementation law**

$$\tilde{a}_t^i \sim \mathbb{P}_{I,i}(\cdot \mid a_t^{i*}, X_t).$$

In the base logit specification (**??**), $\mathbb{P}_{I,i}(\cdot \mid a_t^{i*}, X_t)$ depends on X_t only through μ_t (via the temperature T_t); conditioning on h_t is reserved for extended specifications. The stochastic structure is formalized on a measurable space $(\Omega, \mathcal{F})$. The measurable function $\mathcal{H}_t^i : \mathcal{A}^i \times \Omega \rightarrow$

$\mathcal{A}^i$ materializes the sequential draw: $\tilde{a}_t^i(\omega) = \mathcal{H}_t^i(a_t^{i*}, \omega)$. The **aggregate profile of executed actions** is

$$\tilde{A}_t = (\tilde{a}_t^K, \tilde{a}_t^S, \tilde{a}_t^F). \quad (\text{S3.2})$$

The following regularity assumptions are imposed.

- (i) **Plan dominance and error support.** In the **base specification** with $\underline{c} > 0$, for all $t < \tau$ (with τ as in (??)), $\mathbb{P}_{I,i}$ satisfies strict **plan dominance**, $\mathbb{P}_{I,i}(a_t^{i*} | a_t^{i*}, X_t) > \mathbb{P}_{I,i}(a | a_t^{i*}, X_t)$ for all $a \neq a_t^{i*}$, and **full support**, $\mathbb{P}_{I,i}(\tilde{a}_t^i = a | a_t^{i*}, X_t) > 0$ for all $a \in \mathcal{A}^i$.
- (ii) **Bounded institutional maturation.** The informational component $[H(\mu_t)/H(\mu_0)]e^{-\eta_{\text{cal}} t}$ declines as posterior uncertainty falls, while the structural floor $\underline{c}$ persists in (??). When $\underline{c} > 0$, even if $H(\mu_t) \rightarrow 0$ one has $T_t \geq T_0 \underline{c}$ and $\mathbb{P}_{I,i}(\tilde{a}_t^i = a_t^{i*} | a_t^{i*}, X_t) < 1$, so full support over $\mathcal{A}^i$ is preserved in the long run.

The implementation law adopts the **hybrid-temperature logit** structure:

$$\mathbb{P}_{I,i}(\tilde{a}_t^i = a | a_t^{i*}, X_t) = \frac{\exp(\mathbf{1}\{a = a_t^{i*}\}/T_t)}{\sum_{a' \in \mathcal{A}^i} \exp(\mathbf{1}\{a' = a_t^{i*}\}/T_t)}, \quad (\text{S3.3})$$

with hybrid temperature

$$T_t = T_0 \max\left\{\frac{H(\mu_t)}{H(\mu_0)} e^{-\eta_{\text{cal}} t}, \underline{c}\right\}, \quad (\text{S3.4})$$

where $T_0 > 0$ is the initial temperature, $\eta_{\text{cal}} > 0$ is the calendar decay rate, Shannon entropy is $H(\mu_t) = -\sum_{\theta} \mu_t(\theta) \ln \mu_t(\theta)$ with $H(\mu_0) > 0$ by the full-support prior in Definition ??, and $\underline{c} \in [0, 1]$ is the floor of structural noise. □

REMARK S3.2—Residual noise and the limit $\underline{c} = 0$: The special case $\underline{c} = 0$ is interpreted as a limit without residual noise: it allows $\mathbb{P}_E(\tilde{a}_t^i = a_t^{i*} | \mathcal{F}_t^{\text{pub}}) \rightarrow 1$ when $H(\mu_t) \rightarrow 0$, but sacrifices the persistent full support maintained under $\underline{c} > 0$. The base version with $\underline{c} > 0$ aligns with the absolute continuity or “grain of truth” condition of ?, according to which rational learning must not assign zero probability to events that can occur.

REMARK S3.3—Axiomatic justification of the logit as the unique implementation law: Equation (??) is the *unique* distribution over $\mathcal{A}^i$ that simultaneously satisfies the three axioms:

- (A1) **Full support:** $\mathbb{P}_{I,i}(a | a_t^{i*}, X_t) > 0$ for all $a \in \mathcal{A}^i$.
- (A2) **Dominance:** $\mathbb{P}_{I,i}(a_t^{i*} | a_t^{i*}, X_t) > \mathbb{P}_{I,i}(a | a_t^{i*}, X_t)$ for all $a \neq a_t^{i*}$.
- (A3) **Maximum entropy:** Among all distributions satisfying (A1) and (A2), $\mathbb{P}_{I,i}$ maximizes the Shannon entropy $H = -\sum_a p_a \ln p_a$ subject to the optimal action receiving fixed probability $1 - \varepsilon_t \in (0, 1)$, where $\varepsilon_t := 1 - \mathbb{P}_{I,i}(a_t^{i*} | a_t^{i*}, X_t)$ is the implementation-error rate induced by the logit.

In the parametrization (??), ε_t is decreasing in $1/T_t$. By the characterization theorem of ?, the unique maximizer is the logit in (??). The MDG process is thus axiomatically justified as the unique implementation process compatible with bounded rationality and maximum entropy.

DEFINITION S3.4—Law of the physical outcome: The **physical outcome** m_t is an observable categorical variable on the support $\mathcal{M}_{\text{out}}$, where $\mathcal{M}_{\text{cont}} = \{\text{cont}\}$, $\mathcal{M}_{\text{term}} =$

$\{\text{pay, res, rel, kill}\}$ and $\mathcal{M}_{\text{out}} = \mathcal{M}_{\text{cont}} \cup \mathcal{M}_{\text{term}}$. The label cont represents the continuation of captivity and $j \in \{1, 2, 3, 4\}$ closure by competitive cause j . For $j \in \{1, 2, 3\}$, the effective intensity $\tilde{\lambda}_j(t \mid \theta_K)$ carries the type-specific parameter $\beta_{K,j}(\theta_K)$ in the Cox specification; cause $j = 4$ enters only through the baseline hazard (there is no $\beta_{K,4}$). Conditional on the captor's type and the material state of the period

$$\mathcal{C}_t(\theta_K) = (\theta_K, \theta_F, \theta_V, \theta_S, z, \tilde{a}_t^F, \tilde{a}_t^K, \tilde{a}_t^S, \alpha_t^*, \gamma_t^*, p_{\text{det},t}),$$

the distribution of m_t is defined from the effective intensities $\tilde{\lambda}_j(t \mid \theta_K)$. Denote by

$$Q_t(m \mid \theta_K, \mathcal{C}_t, \alpha_t^*, \gamma_t^*) := \mathbb{P}_{\text{E}}(m_t = m \mid \theta_K, \mathcal{C}_t, \alpha_t^*, \gamma_t^*)$$

the transition kernel of the physical outcome, and by G_t the measurable rule that materializes a concrete realization of m_t from that kernel and a uniform draw. The **continuation probability** is

$$Q_t(\text{cont} \mid \theta_K, \mathcal{C}_t, \alpha_t^*, \gamma_t^*) = \exp\left(-\sum_{j=1}^4 \tilde{\lambda}_j(t \mid \theta_K) \Delta t\right).$$

The daily closure total and the relative weight of each cause are

$$q_t(\theta_K) = 1 - p_{\text{Cont},t}, \quad \xi_j = \frac{\tilde{\lambda}_j}{\sum_{\ell} \tilde{\lambda}_{\ell}}, \quad \bar{h}_j(t \mid \theta_K) = q_t \xi_j.$$

The materialization of m_t is obtained by **inverse transform**: let $v_t \sim \text{Unif}(0, 1)$; then

$$m_t = r \iff \sum_{\ell < r} Q_t(\ell \mid \theta_K, \mathcal{C}_t, \alpha_t^*, \gamma_t^*) \leq v_t < \sum_{\ell \leq r} Q_t(\ell \mid \theta_K, \mathcal{C}_t, \alpha_t^*, \gamma_t^*),$$

and the likelihood of m_t in the Bayesian update is

$$\mathbb{P}_{\text{E}}(m_t \mid \theta_K, \mathcal{C}_t) = p_{\text{Cont}}^{\mathbf{1}\{m_t=\text{cont}\}} \prod_{j=1}^4 \bar{h}_j^{\mathbf{1}\{m_t=j\}}.$$

□

S4. ASYMPTOTIC TYPE IDENTIFICATION

This section defines the prolonged captivity event and the notion of asymptotic identification, states the structural identifiability lemma (bridge between the parameters $\beta_{K,j}$ and the singularity of laws), and concludes with the posterior consistency theorem.

Notation. Throughout Sections ?? and ??, θ_K^* denotes the realized captor type. When conditioning on the law of the true type, the notation $\theta^* := \theta_K^*$ and $\mu_C := \mathbb{P}_{C,\theta^*}(\cdot \mid \mathcal{T})$ is adopted.

DEFINITION S4.1—Prolonged captivity event: Let τ be as in (??). The **prolonged captivity event** is

$$\mathcal{T} := \{\tau = \infty\} = \{\omega \in \Omega : m_t(\omega) \in \mathcal{M}_{\text{cont}} \forall t \geq 0\}.$$

In $\mathcal{T}$ the episode admits no terminal outcome in any finite period and generates an infinite sequence of public signals $\{\zeta_t\}_{t \geq 0}$. □

DEFINITION S4.2—Asymptotic identification in the event $\mathcal{T}$: The model is **asymptotically identifiable** in $\mathcal{T}$ (Definition ??) if $\mathbb{P}_{C, \theta_K^*}(\mathcal{T}) > 0$ and, for every $\theta \in \Theta_K$ with $\theta \neq \theta_K^*$, at least one of the following conditions holds.

- (a) **Exclusion by survival.** $\mathbb{P}_{C, \theta}(\mathcal{T}) = 0$.
- (b) **Tail singularity of signals.** $\mathbb{P}_{C, \theta}(\mathcal{T}) > 0$ and the laws of $\{\zeta_t\}_{t \geq 0}$ restricted to $\mathcal{T}$ under $\mathbb{P}_{C, \theta}$ and under $\mathbb{P}_{C, \theta_K^*}$ are **mutually singular**. □

REMARK S4.3—Role of Lemma ?? in the analysis: Theorem ?? requires asymptotic identification (Definition ??), a condition formulated in terms of laws of entire processes. Lemma ?? is the **operational bridge** between that abstract hypothesis and the parameters $\beta_{K,j}$ already introduced in the mechanism: it allows one to verify in the practice of the model when two distinct types generate asymptotically distinguishable trajectories. Without this result, the learning section would rest on a high-level condition difficult to connect with the proportional parametrization of the λ_j .

LEMMA S4.4—Structural identifiability and asymptotic learning: Suppose $|\Theta_K| < \infty$. Let $Y_t := (m_t, d_t)$ be the daily public record. Under a PBE, $p_\theta(\cdot \mid \mathcal{F}_t^{\text{pub}})$ denotes the mass function of Y_t conditioned on public information at the opening of t under type θ .

Part (i) — local sufficiency (??). Let $\theta, \theta' \in \Theta_K$ be distinct. If there exists $j \in \{1, 2, 3\}$ such that $\beta_{K,j}(\theta) \neq \beta_{K,j}(\theta')$ (causes $j \in \{1, 2, 3\}$ only; see Definition ??) and no exact compensations arise between the masses of Y_t (knife-edges), then there exists $\varepsilon = \varepsilon(\theta, \theta', t) > 0$ such that

$$\|p_\theta(\cdot \mid \mathcal{F}_t^{\text{pub}}) - p_{\theta'}(\cdot \mid \mathcal{F}_t^{\text{pub}})\|_{\text{TV}} \geq \varepsilon. \quad (\text{S4.1})$$

Part (ii) — asymptotic sufficiency (??). Fixing a policy trajectory $(\chi_s)_{s \geq 0}$, suppose that conditional on $\mathcal{T}$ the $\{Y_t\}_{t \geq 0}$ are independent. For each pair $\theta \neq \theta'$, let $\mathcal{S}_{\theta, \theta'} \subseteq \mathbb{N}$ be the set of periods where (??) holds. If $\mathcal{S}_{\theta, \theta_K^*}$ is infinite $\mathbb{P}_{C, \theta_K^*}$ -a.s. and there exists $\varepsilon_\theta > 0$ such that $\|p_\theta^{(t)} - p_{\theta_K^*}^{(t)}\|_{\text{TV}} \geq \varepsilon_\theta$ for all $t \in \mathcal{S}_{\theta, \theta_K^*}$, then the laws of $(Y_t)_{t \geq 0}$ conditioned on $\mathcal{T}$ under $\mathbb{P}_{C, \theta}$ and under $\mathbb{P}_{C, \theta_K^*}$ are **mutually singular**.

PROOF: The proof identifies the causal mechanism within the day, translates it to a statistical bound in total variation, and couples the repeated experiment to the criterion of ?.

1. Same control menu, distinct criminal technology. Fix t and $\mathcal{F}_t^{\text{pub}}$. In the counterfactual that changes only the type (θ vs. θ'), keeping equal the executed triple $\tilde{A}_t$, the instruments (α_t^*, γ_t^*) , and the maturation filter $M(t)$, the proportional parametrization implies

$$\frac{\lambda_j(t \mid \theta)}{\lambda_j(t \mid \theta')} = \exp(\beta_{K,j}(\theta) - \beta_{K,j}(\theta')) \neq 1.$$

2. From daily intensity to distinct predictions over Y_t . The proportional difference in $\lambda_j(t)$ is projected onto the aggregate exponential of $q(t)$, the causal mix $\xi_j(t)$, or the marginal hazards $h_j(t) = q(t)\xi_j(t)$. Except for *knife-edges*, the marginal of m_t cannot coincide under θ and θ' , and the two mass functions differ in total variation, verifying (??).

3. Repetition with uniform margin and Kakutani criterion. Conditional on the fixed policy and on $\mathcal{T}$, the law of the sequence is a product of daily experiments $\bigotimes_{t \geq 0} p_\theta^{(t)}$ versus $\bigotimes_{t \geq 0} p_{\theta_K^*}^{(t)}$. Part (ii) then follows from Lemma ?? (Section ??): recurrent uniform total-variation separation on an infinite set forces the Bhattacharyya product affinity to zero and, by the criterion of ?, mutual singularity of the infinite product measures (?). Q.E.D.

THEOREM S4.5—Identification and consistency of true types: *Consider a prior $\mu_0 \in \Delta(\Theta_K)$ with full support and a PBE-consistent Bayesian updating (Definition ??). Let $\theta^* := \theta_K^*$ be the realized type and $\mu_C := \mathbb{P}_{C, \theta^*}(\cdot \mid \mathcal{T})$ the law of the true type restricted to the event $\mathcal{T}$ (Definition ??).*

If the mechanism is asymptotically identifiable in $\mathcal{T}$ (Definition ??), the following properties are satisfied μ_C -almost surely:

1. **Exclusion of false hypotheses:** $\mu_t(\theta) \xrightarrow{t \rightarrow \infty} 0$ for all $\theta \neq \theta^*$.
2. **Concentration on the true type:** $\mu_t(\theta^*) \xrightarrow{t \rightarrow \infty} 1$.

Consequently, in $\mathcal{T}$ the sequence $\{\mu_t\}$ converges weakly to the Dirac measure δ_{θ^} .*

PROOF: The proof separates the general probabilistic argument from the structural identification assumption.

S4.0.0.1. 1. Stability of beliefs. By equilibrium consistency, $\mu_t(\theta) = \mathbb{E}^{\mathbb{P}_E}[\mathbf{1}\{\theta_K = \theta\} \mid \mathcal{F}_t^{\text{pub}}] \mathbb{P}_E\text{-a.s.}$ Lemma ?? guarantees that $\{\mu_t(\theta)\}$ is a bounded martingale with limit $Z_\theta \in [0, 1] \mathbb{P}_E\text{-a.s.}$ The transfer to $\mu_C\text{-a.s.}$ is justified by disintegration: the mixture $\mathbb{P}_E = \sum_{\vartheta} \mu_0(\vartheta) \mathbb{P}_{C, \vartheta}$ implies that every $\mathbb{P}_E$ -null set is $\mathbb{P}_{C, \theta^*}$ -null (since $\mu_0(\theta^*) > 0$), and domination is preserved upon restricting to $\mathcal{T}$.

S4.0.0.2. 2. Identification and exclusion of false types. Fix $\theta \neq \theta^*$.

Case $\mathbb{P}_{C, \theta}(\mathcal{T}) = 0$: the trajectory under μ_C belongs to an event that the false type cannot generate; its posterior likelihood is null in the conditioned regime.

Case $\mathbb{P}_{C, \theta}(\mathcal{T}) > 0$: the singularity of Definition ??(b) implies that there exists a tail event A_θ with $\mathbb{P}_{C, \theta^*}(A_\theta \mid \mathcal{T}) = 1$ and $\mathbb{P}_{C, \theta}(A_\theta \mid \mathcal{T}) = 0$. Let R_t^θ be the likelihood ratio of the false type against the true type on finite horizons. Tail singularity forces $R_t^\theta \rightarrow 0$ $\mu_C\text{-a.s.}$ (??). By Bayes' formula with a full-support prior,

$$\mu_t(\theta) = \frac{\mu_0(\theta) R_t^\theta}{\mu_0(\theta^*) + \sum_{\theta' \neq \theta^*} \mu_0(\theta') R_t^{\theta'}} \leq \frac{\mu_0(\theta)}{\mu_0(\theta^*)} R_t^\theta,$$

and since $\mu_0(\theta^*) > 0$, it follows that $\mu_t(\theta) \rightarrow 0$ $\mu_C\text{-a.s.}$ for all $\theta \neq \theta^*$.

S4.0.0.3. 3. *Closure by normalization.* For all t , $\sum_{\theta \in \Theta_K} \mu_t(\theta) = 1$. As Θ_K is finite and all false coordinates converge to zero μ_C -a.s.,

$$1 = \lim_{t \rightarrow \infty} \sum_{\theta} \mu_t(\theta) = Z_{\theta^*} + \sum_{\theta \neq \theta^*} 0,$$

hence $Z_{\theta^*} = 1$ μ_C -a.s., that is, $\mu_t(\theta^*) \rightarrow 1$ μ_C -a.s. On a finite simplex, coordinate convergence is equivalent to weak convergence $\mu_t \rightarrow \delta_{\theta^*}$. *Q.E.D.*

REMARK S4.6—Economic reading and link with $\beta_{K,j}$: Definition ?? requires that the design of the game (policy + incentives + risk technology) produce long trajectories incompatible with false types. Distinct shifts $\beta_{K,j}(\theta) \neq \beta_{K,j}(\theta')$ favor the singularity of the laws of (m_t, d_t) . If the equilibrium pools types, Lemma ?? still holds but Theorem ?? does not: the limit may split mass among observationally equivalent types.

REMARK S4.7—Finite horizon: If $\tau < \infty$ almost surely, $\mathcal{T}$ has measure zero and the statement of Theorem ?? is vacuous. The relevant conclusion is then only the martingale convergence of Lemma ??: beliefs stabilize at the finite limit of observations.

S5. COMPETITIVE RISKS AND OPERATIONAL PRESSURE

The following proposition uses directly the proportional risk structure, without requiring the equilibrium machinery of Section ?. It depends only on the intensities $\tilde{\lambda}_j$ introduced in Definition ??.

PROPOSITION S5.1—Expected duration of captivity and operational pressure: *Fix a type θ_K and a period t . The following standing assumptions govern the comparative static in γ_t^* .*

Standing assumptions. (S0) *Baseline hazards satisfy $\lambda_{j0}(s) > 0$ for all strategic causes $j \in \{1, 2, 3\}$ and all s in the support, so $\mathbb{E}[\tau | \theta_K] < \infty$.* (S1) *The object $\partial \mathbb{E}[\tau | \theta_K] / \partial \gamma_t^*$ is a partial derivative: only γ_t^* is perturbed at period t while $\{\gamma_s^*\}_{s \neq t}$ is held fixed. Writing $S(s | \theta_K) := \prod_{r=1}^s p_{\text{Cont}, r | \theta_K}$ with $S(0 | \theta_K) = 1$, expected duration admits the survival representation*

$$\mathbb{E}[\tau | \theta_K] = \sum_{s=0}^{\infty} S(s | \theta_K), \quad (\text{S5.1})$$

so only factors with $s \geq t$ depend on γ_t^ .* (S2) *For each strategic cause $j \in \{1, 2, 3\}$,*

$$\frac{\partial \tilde{\lambda}_j(t | \theta_K)}{\partial \gamma_t^*} = \eta_{\gamma,j} \tilde{\lambda}_j(t | \theta_K), \quad \eta_{\gamma,j} := \frac{\partial \ln \tilde{\lambda}_j(t | \theta_K)}{\partial \gamma_t^*};$$

the exogenous cause $j = 4$ is invariant to γ_t^ .*

Net sensitivity. *In the main-paper hazard index, payment ($j = 1$) is decreasing in γ and death/state rescue ($j \in \{2, 3\}$) are increasing, so $\eta_{\gamma,1} < 0$ and $\eta_{\gamma,j} > 0$ for $j \in \{2, 3\}$. Let $\zeta_{\gamma,j} := |\eta_{\gamma,j}|$ and define*

$$\kappa_h(\theta_K, t) := \zeta_{\gamma,2} \tilde{\lambda}_2(t | \theta_K) + \zeta_{\gamma,3} \tilde{\lambda}_3(t | \theta_K) - \zeta_{\gamma,1} \tilde{\lambda}_1(t | \theta_K) = \sum_{j=1}^3 \eta_{\gamma,j} \tilde{\lambda}_j(t | \theta_K). \quad (\text{S5.2})$$

Result. Under (S0)–(S2),

$$\frac{\partial \mathbb{E}[\tau \mid \theta_K]}{\partial \gamma_t^*} = -\kappa_h(\theta_K, t) \Delta t \sum_{s=t}^{\infty} S(s \mid \theta_K). \quad (\text{S5.3})$$

In particular, whenever $\kappa_h(\theta_K, t) \neq 0$,

$$\text{sgn}\left(\frac{\partial \mathbb{E}[\tau \mid \theta_K]}{\partial \gamma_t^*}\right) = -\text{sgn}(\kappa_h(\theta_K, t)). \quad (\text{S5.4})$$

If $\kappa_h(\theta_K, t) > 0$, the one-shot perturbation in (S1) lowers expected duration; if $\kappa_h(\theta_K, t) < 0$, it raises it. Cross-type variation in the sign enters through $\beta_{K,j}(\theta_K)$ and the induced semi-elasticities $\{\eta_{\gamma,j}\}$.

PROOF: The argument has three steps: differentiate the daily continuation probability, propagate the perturbation through the survival product, and differentiate the duration identity (??).

S5.0.0.1. Step 0. Survival representation. For a discrete-time competing-risks process with daily continuation probability $p_{\text{Cont},s|\theta_K}$, the survival function $S(t' \mid \theta_K) = \prod_{s=1}^{t'} p_{\text{Cont},s|\theta_K}$ satisfies $S(t' \mid \theta_K) = \mathbb{P}(\tau > t' \mid \theta_K)$. Hence

$$\mathbb{E}[\tau \mid \theta_K] = \sum_{t'=0}^{\infty} \mathbb{P}(\tau > t' \mid \theta_K) = \sum_{t'=0}^{\infty} S(t' \mid \theta_K),$$

which is the discrete-time survival identity (Wald's representation; see, e.g., ?). Finiteness in (S0) makes the sum well defined.

S5.0.0.2. Step 1. Daily continuation and κ_h . By Definition ??,

$$p_{\text{Cont},s|\theta_K} = \exp\left(-\sum_{j=1}^4 \tilde{\lambda}_j(s \mid \theta_K) \Delta t\right) = \exp(-[\Lambda_s(\theta_K) + \tilde{\lambda}_4(s)] \Delta t),$$

where $\Lambda_s(\theta_K) := \sum_{j=1}^3 \tilde{\lambda}_j(s \mid \theta_K)$ is the strategic hazard total. Only Λ_t depends on γ_t^* ; the exogenous term $\tilde{\lambda}_4(t)$ is γ -invariant. By (S2),

$$\frac{\partial \Lambda_t(\theta_K)}{\partial \gamma_t^*} = \sum_{j=1}^3 \eta_{\gamma,j} \tilde{\lambda}_j(t \mid \theta_K) = \kappa_h(\theta_K, t),$$

using $\eta_{\gamma,1} = -\zeta_{\gamma,1}$ and $\eta_{\gamma,j} = +\zeta_{\gamma,j}$ for $j \in \{2, 3\}$ in (??). Since $p_{\text{Cont},t|\theta_K} = \exp(-\Lambda_t \Delta t) \cdot \exp(-\tilde{\lambda}_4(t) \Delta t)$ and the second factor is constant in γ_t^* ,

$$\frac{\partial p_{\text{Cont},t|\theta_K}}{\partial \gamma_t^*} = \exp(-\tilde{\lambda}_4(t) \Delta t) \cdot \frac{\partial}{\partial \gamma_t^*} \exp(-\Lambda_t \Delta t) = -\kappa_h(\theta_K, t) p_{\text{Cont},t|\theta_K} \Delta t. \quad (\text{S5.5})$$

S5.0.0.3. Step 2. Propagation to accumulated survival. Fix $t' \geq t$. Because only the factor $s = t$ in $S(t' \mid \theta_K) = \prod_{s=1}^{t'} p_{\text{Cont},s|\theta_K}$ depends on γ_t^* ,

$$\frac{\partial S(t' \mid \theta_K)}{\partial \gamma_t^*} = \left(\prod_{\substack{s=1 \\ s \neq t}}^{t'} p_{\text{Cont},s|\theta_K} \right) \frac{\partial p_{\text{Cont},t|\theta_K}}{\partial \gamma_t^*} = -\kappa_h(\theta_K, t) S(t' \mid \theta_K) \Delta t.$$

For $t' < t$, the derivative is zero.

S5.0.0.4. Step 3. Expected duration and sign. Differentiating (??) term by term under (S1) and applying Step 2 yields (??). The interchange of sum and derivative is justified by dominated convergence: for $t' \geq t$, $|\partial S(t' | \theta_K) / \partial \gamma_t^*| \leq |\kappa_h(\theta_K, t)| S(t' | \theta_K) \Delta t$, and

$$\sum_{t'=t}^{\infty} S(t' | \theta_K) \leq \sum_{t'=0}^{\infty} S(t' | \theta_K) = \mathbb{E}[\tau | \theta_K] < \infty$$

by (S0). Because $S(t' | \theta_K) \geq 0$ and the tail $\sum_{t'=t}^{\infty} S(t' | \theta_K) > 0$ whenever continuation has positive mass on $t' \geq t$, the sign of (??) is the opposite of the sign of $\kappa_h(\theta_K, t)$. *Q.E.D.*

REMARK S5.2—Local comparative static and testable content: Proposition ?? is a *local* comparative static: it evaluates a one-period perturbation of γ_t^* holding the remaining policy path fixed. It does not claim global monotonicity of $\mathbb{E}[\tau | \theta_K]$ as a function of the entire sequence $\{\gamma_s^*\}_{s \geq 0}$. The empirical content is the sign prediction: types with larger κ_h respond differently to operational pressure because the Cox coefficients $\beta_{K,j}(\theta_K)$ shift the semi-elasticities $\{\eta_{\gamma,j}\}$ and the levels $\{\tilde{\lambda}_j\}$.

S6. POLICY IMPLICATIONS

This proposition requires the belief machinery of Section ?? and the equilibrium context of Section ??: optimal pressure is calibrated in a Bayesian manner.

PROPOSITION S6.1—Optimal operational pressure and Bayesian type identification: Suppose $\Theta_K = \{\theta_{low}, \theta_{high}\}$ and fix a discrete branch $b \in \{R, N\}$ in a neighborhood of the posterior belief under consideration. Let $x_t = (\alpha_t, \gamma_t)$ and define the complete branch objective

$$J_t^b(x_t, \mu_t) := \sum_{\theta \in \Theta_K} \mu_t(\theta) L_t(b, \alpha_t, \gamma_t, \theta, \iota_t) + \Pi_{t,b}^S(\alpha_t, \gamma_t; \mu_t) - \psi_H \Delta H_t(\alpha_t, \gamma_t). \quad (\text{S6.1})$$

If (α_t^*, γ_t^*) is an interior optimum of $J_t^b(\cdot, \mu_t)$, the constraints IC^K , IR^K , and IR^F do not locally change regime, the Hessian $H_t^b := \nabla_{xx}^2 J_t^b(x_t^*, \mu_t)$ is positive definite, and the net marginal effect of increasing $\mu_t(\theta_{high})$ on the first-order condition for γ_t favors greater pressure, that is,

$$e'_\gamma (H_t^b)^{-1} \nabla_{x\mu}^2 J_t^b(x_t^*, \mu_t) < 0, \quad e_\gamma = (0, 1)', \quad (\text{S6.2})$$

then the optimal operational pressure of that branch is locally increasing in the posterior belief on the high-risk type:

$$\frac{d\gamma_t^*}{d\mu_t(\theta_{high})} > 0.$$

The structural separation $\beta_{K,j}(\theta_{high}) - \beta_{K,j}(\theta_{low})$ for $j \in \{2, 3\}$ contributes to this condition through the ratio

$$\frac{\tilde{\lambda}_j(t | \theta_{high})}{\tilde{\lambda}_j(t | \theta_{low})} = \exp(\beta_{K,j}(\theta_{high}) - \beta_{K,j}(\theta_{low})), \quad j \in \{2, 3\}.$$

Monotonicity, however, depends on the joint net effect of material losses, deviation penalties, and informational gains.

PROOF: Write $\mu := \mu_t(\theta_{high})$ and $\mu_t(\theta_{low}) = 1 - \mu$. Within the fixed discrete branch b , the interior optimum $x_t^*(\mu) = (\alpha_t^*(\mu), \gamma_t^*(\mu))$ satisfies

$$\nabla_x J_t^b(x_t^*(\mu), \mu) = 0. \quad (S6.3)$$

The hypothesis $H_t^b = \nabla_{xx}^2 J_t^b(x_t^*, \mu) \succ 0$ implies that the local optimum is isolated and that the Jacobian of the first-order condition with respect to x_t is invertible. By the implicit function theorem, $x_t^*(\mu)$ is locally differentiable and

$$\frac{dx_t^*}{d\mu} = -(H_t^b)^{-1} \nabla_{x\mu}^2 J_t^b(x_t^*, \mu). \quad (S6.4)$$

Taking the second coordinate yields

$$\frac{d\gamma_t^*}{d\mu_t(\theta_{high})} = -e'_\gamma(H_t^b)^{-1} \nabla_{x\mu}^2 J_t^b(x_t^*, \mu_t). \quad (S6.5)$$

By the net monotonicity condition (??), the right-hand side of (??) is strictly positive.

To interpret this condition, note that the coordinate of $\nabla_{x\mu}^2 J_t^b$ associated with γ_t contains

$$\frac{\partial L_t}{\partial \gamma_t}(b, \alpha_t^*, \gamma_t^*, \theta_{high}, \mu_t) - \frac{\partial L_t}{\partial \gamma_t}(b, \alpha_t^*, \gamma_t^*, \theta_{low}, \mu_t) + \frac{\partial}{\partial \mu} \left[\frac{\partial \Pi_{t,b}^S}{\partial \gamma_t} - \psi_H \frac{\partial \Delta H_t}{\partial \gamma_t} \right]_{x_t=x_t^*}. \quad (S6.6)$$

The first differential captures the material separation between types. Under proportional risks,

$$\frac{\tilde{\lambda}_j(t | \theta_{high})}{\tilde{\lambda}_j(t | \theta_{low})} = \exp(\beta_{K,j}(\theta_{high}) - \beta_{K,j}(\theta_{low})),$$

so that greater structural distance in $j \in \{2, 3\}$ increases the difference in marginal losses associated with operational pressure. However, the deviation premium and the informational subsidy also enter the first-order condition; therefore the result is formulated as monotonicity of the net marginal effect of the complete objective, not as a property of material loss alone. Under (??), increasing the posterior mass on the high-risk type shifts the interior optimum toward a higher γ_t^* . Q.E.D.

S7. EQUILIBRIUM AND IMPLEMENTABILITY

This section introduces the dynamic mechanism with its PBE, the notion of implementability with the constraints IC^K , IR^K , IR^F , and proves the existence of the implementable equilibrium.

DEFINITION S7.1—Dynamic mechanism and Perfect Bayesian Equilibrium: Consider a discrete-time Bayesian game with horizon $\mathbb{T} = \{0, 1, \dots, \tau\}$, players $\mathcal{J} = \{S, K, F\}$, and captor type $\theta_K \in \Theta_K$ according to common prior μ_0 with $\mu_0(\theta) > 0$ for all θ . A **dynamic mechanism** is described by:

- *Strategy profile:* $s = (s^S, s^K, s^F)$, where $s^S = \chi = (\chi_t)_{t \in \mathbb{T}}$ with $\chi_t : H_t \times \mathcal{R} \rightarrow \mathcal{A}^{S, \text{disc}} \times \mathcal{A}^\alpha \times \mathcal{A}^\gamma$.
- *Belief system:* $\mu = \{\mu_t\}_{t \geq 0}$, $\mu_t \in \Delta(\Theta_K)$.
- *Ex-ante law:* $\mathbb{P}_E^s$ induced by profile s , prior μ_0 , and the physical dynamics of captivity.

Perfect Bayesian Equilibrium. The pair (s, μ) is a **PBE** if the following hold:

- (a) **Sequential rationality.** At every node reached with positive probability, each agent i solves its optimization program given μ_t and s^{-i} . For the state:

$$(a_t^{S*}, \alpha_t^*, \gamma_t^*) \in \arg \min_{(a_t^S, \alpha_t, \gamma_t)} \left\{ \sum_{\theta} \mu_t(\theta) L_t(a_t^S, \alpha_t, \gamma_t, \theta, \iota_t) + \Pi_{t,a^S}^S(\alpha_t, \gamma_t; \mu_t) - \psi_H \Delta H_t(\alpha_t, \gamma_t) \right\}.$$

For $i \in \{K, F\}$: $a_t^{i*} \in \arg \max_{a_t^i} \mathcal{U}_t^i(a_t^i | \mathcal{I}_t^i, \chi, s^{-i})$.

- (b) **On-path Bayesian consistency.**

$$\mu_t(\theta) = \mathbb{P}_E^s(\theta_K = \theta | \mathcal{F}_t^{\text{pub}}), \quad \forall \theta, t \geq 1.$$

□

REMARK S7.2: Definition ?? defines the PBE as an *object*: the pair (s, μ) satisfying (a) and (b). It does not assert existence nor that the profile satisfies incentive and participation constraints; those properties are the content of Theorem ??.

DEFINITION S7.3—Nash-Bayes- IC^K, IR^K, IR^F implementability: Let (s^*, μ^*) be a PBE (Definition ??). The mechanism is **implementable Nash-Bayes- IC^K, IR^K, IR^F** if the profile additionally satisfies the following conditions:

Bayesian incentive compatibility of the captor (IC^K). For all $t \in \mathbb{T}$ and all rival mimicry $\theta_j \in \Theta_K$,

$$\mathbb{E}_{\theta \sim \mu_t^*} [V_t^K(a^*(\theta) | \theta, \alpha_t^*, \gamma_t^*) - V_t^K(a^*(\theta_j) | \theta, \alpha_t^*, \gamma_t^*)] \geq 0.$$

Participation constraint of the captor (IR^K).

$$\mathbb{E}_{\theta \sim \mu_t^*} [U_{\text{rel}}^K(\theta) - \max\{V_{\text{cont},t}^K(\theta, \alpha_t^*, \gamma_t^*), U_{\text{kill}}^K(\theta, \theta_S)\}] \geq 0.$$

Participation constraint of the family (IR^F).

$$\mathcal{U}_t^F(a_{\text{coop}}) \geq \mathcal{U}_t^F(a_{\text{col}}).$$

The **feasible set of policies** is

$$\Gamma_t(\mu_t) := \left\{ (a_t^S, \alpha_t, \gamma_t) \in \mathcal{A}^{S, \text{disc}} \times [0, 1]^2 : IC^K, IR^K, IR^F \text{ are satisfied} \right\}. \quad (\text{S7.1})$$

Its non-emptiness — **robust feasibility** — is the primitive hypothesis of Theorem ??.

□

REMARK S7.4: Definition ?? states implementability as a *property* of the pair (s^*, μ^*) : given the existence of the PBE, one asks whether it additionally satisfies IC^K , IR^K , and IR^F . The cycle criterion of ? provides a useful algebraic reformulation for auditing IC^K when Θ_K is finite.

REMARK S7.5—Full support of the MDG and Bayesian operability: The full support of $\mathbb{P}_{I,i}$ (hypothesis (iii) of Theorem ??) guarantees strictly positive probability at all reachable nodes, making Bayesian updating operative on the path and reducing indeterminacy off the path.

THEOREM S7.6—Conditional implementability of the mechanism in PBE: *Consider the mechanism of Definition ?? under the following conditions:*

- (i) $|\Theta_K| < \infty$ with strictly positive prior.
- (ii) Finite action spaces $\mathcal{A}^i$ for all $i \in \{K, S, F\}$.
- (iii) The law $\mathbb{P}_{I,i}$ of the MDG process assigns strictly positive probability to every action (**full support**).
- (iv) $\mathcal{A}^\alpha = \mathcal{A}^\gamma = [0, 1]$, compact and convex; L_t and $\mathcal{U}_t^i$ are measurable, bounded, and continuous in the continuous decision variables. The physical outcome rule G_t is measurable, and the transition kernel Q_t induced by G_t and by the competitive risk probabilities is continuous in those variables.
- (v) **Robust feasibility.** For each t and $\mu_t \in \Delta(\Theta_K)$, $\Gamma_t(\mu_t)$ is non-empty and compact; the sections $\Gamma_t^{a^S}(\mu_t)$ are convex; the correspondence $\mu_t \mapsto \Gamma_t(\mu_t)$ is continuous with compact values.
- (vi) **Parametric separability.**

$$V_t^K(a^*(\theta') \mid \theta_K, \alpha_t, \gamma_t) = \sum_j w_j(a^*(\theta'), \alpha_t, \gamma_t) \exp(\beta_{K,j}(\theta_K) \cdot z).$$

- (vii) **Quasiconvexity of the restricted objective.** For each t , μ_t , and a_t^S , the functional $(\alpha_t, \gamma_t) \mapsto \sum_\theta \mu_t(\theta) L_t + \Pi_{t,a^S}^S - \psi_H \Delta H_t$ is quasiconvex in (α_t, γ_t) over $\Gamma_t^{a^S}(\mu_t)$.

Then there exists (s^*, μ^*) such that:

- (a) Constitutes a **PBE** (Definition ??).
- (b) **Implements the social choice rule** f_t : $\chi_t^*(h_t) = (a_t^{S*}, \alpha_t^*, \gamma_t^*)$ minimizes the expected social loss subject to $\Gamma_t(\mu_t^*)$.
- (c) **Satisfies** IC^K , IR^K , and IR^F in profile (s^*, μ^*) .

PROOF: The proof proceeds in five steps.

S7.0.0.1. 1. *Existence of equilibrium.* By (ii), the behavioral strategy space Σ^i of each agent is a finite product of simplices, compact and convex. By (v), $\Gamma_t(\mu_t)$ is compact, non-empty, with convex sections and continuous correspondence. The state's objective is continuous in (α_t, γ_t) by (iv); by **Berge's maximum theorem** (?), the correspondence of optima $\Psi_t(\mu_t)$ is upper hemicontinuous and compact. By (vii), the functional is quasiconvex; the argmin of a quasiconvex function over a compact convex set is convex, so $\Psi_t(\mu_t)$ has convex values. The utilities of K and F are multilinear in behavioral strategies, so the best response correspondence

$$\text{BR}(s) := \text{BR}_t^S(s^{-S}) \times \text{BR}^K(s^{-K}) \times \text{BR}^F(s^{-F})$$

has non-empty convex values. By **Kakutani's fixed-point theorem** (?), there exists s^* with $s^* \in \text{BR}(s^*)$.

S7.0.0.2. 2. *On-path Bayesian consistency.* By (iii), $\mathbb{P}_{I,i}$ assigns positive probability to all actions compatible with the physical dynamics. Bayes' formula

$$\mu_t^*(\theta) = \mathbb{P}_E^S(\theta_K = \theta \mid \sigma(z, \zeta_0, \dots, \zeta_{t-1}))$$

is well defined at reached nodes and satisfies condition (b) of Definition ??.

S7.0.0.3. 3. *Implementation of the social choice rule.* At the fixed point, the state's strategy solves at each node on the path:

$$(a_t^{S*}, \alpha_t^*, \gamma_t^*) \in \arg \min_{(a_t^S, \alpha_t, \gamma_t) \in \Gamma_t(\mu_t^*)} \left\{ \sum_{\theta} \mu_t^*(\theta) L_t + \Pi_{t,a^S}^S(\alpha_t, \gamma_t; \mu_t^*) - \psi_H \Delta H_t(\alpha_t, \gamma_t) \right\}.$$

The objective is continuous by (iv) and $\Gamma_t(\mu_t^*)$ is compact and non-empty by (v); by Weierstrass a solution exists.

S7.0.0.4. 4. *Incentive compatibility (IC^K): cycle reformulation (?).* IC^K is imposed through the feasible set $\Gamma_t(\mu_t^*)$, not verified ex post: equilibrium instruments belong to $\Gamma_t(\mu_t^*)$, which by (v) and Definition ?? incorporates IC^K as a constitutive condition. For algebraic auditing with $\Delta_{ij}^t := V_t^K(a^*(\theta_j) \mid \theta_i) - V_t^K(a^*(\theta_i) \mid \theta_i)$, the Bayesian version is equivalent to $\mathbb{E}_{\theta_i \sim \mu_t^*} [\Delta_{ij}^t] \leq 0$ for all j . For every cycle $(\theta_{k_0}, \dots, \theta_{k_m}, \theta_{k_0})$:

$$\sum_{\ell=0}^m \mu_t^*(\theta_{k_\ell}) \Delta_{k_\ell, k_{\ell+1}}^t \leq 0 \quad (\text{indices mod } m+1).$$

S7.0.0.5. 5. *Participation constraints (IR^K, IR^F).* By construction in Step 3, $(\alpha_t^*, \gamma_t^*) \in \Gamma_t(\mu_t^*)$, a set that by (v) simultaneously guarantees IR^K and IR^F in the equilibrium profile.

Having completed the five steps, (s^*, μ^*) is a PBE that implements f_t and satisfies IC^K , IR^K , and IR^F simultaneously. *Q.E.D.*

REMARK S7.7—Separation of results: implementability and asymptotic identification: Theorem ?? and Theorem ?? guarantee complementary properties. The first guarantees existence of a PBE with implementable policy subject to Γ_t : IC^K enters through the definition of the feasible set. The second guarantees that under that same equilibrium, if heterogeneity in $\beta_{K,j}$ generates uniform separation in total variation, beliefs converge to the true type almost surely in $\mathcal{T}$. The full support of the MDG plays a role in both: it reduces the *off-path* region and avoids degeneracies of the likelihood.

S8. ABSTRACT STRUCTURAL IDENTIFICATION (KAKUTANI CRITERION)

This section isolates the measure-theoretic argument: under product structure and recurrent separation in total variation, the criterion of ? forces mutual singularity of trajectory laws. Theorem ?? states the same Bayesian conclusion as Section ??, replacing Lemma ?? with the abstract hypotheses of the following lemma.

LEMMA S8.1—Product of experiments and singularity (Kakutani criterion): *Fix $\theta \neq \theta'$. Suppose that conditional on equilibrium policy and on event $\mathcal{T}$, the sequence $\{Y_t\}_{t \geq 0}$ is independent and each Y_t takes values in a common finite support $\mathcal{Y}_t$. Let $\rho_t := \rho(p_\theta^{(t)}, p_{\theta'}^{(t)})$ be the*

Bhattacharyya affinity. Then the affinity between the product laws on horizon T is

$$\rho\left(\bigotimes_{t=0}^T p_{\theta}^{(t)}, \bigotimes_{t=0}^T p_{\theta'}^{(t)}\right) = \prod_{t=0}^T \rho_t. \quad (\text{S8.1})$$

*If there exists an infinite set $\mathcal{S}_{\theta, \theta'} \subseteq \mathbb{N}$ with $\|p_{\theta}^{(t)} - p_{\theta'}^{(t)}\|_{\text{TV}} \geq \underline{\varepsilon} > 0$ for all $t \in \mathcal{S}_{\theta, \theta'}$, then $\prod_{t \in \mathcal{S}} \rho_t = 0$ and, by the criterion of ?, the product measures over the infinite sequence are **mutually singular** (?).*

PROOF: Identity (??). Under independence,

$$\rho\left(\bigotimes_{t=0}^T p_{\theta}^{(t)}, \bigotimes_{t=0}^T p_{\theta'}^{(t)}\right) = \sum_{(y_0, \dots, y_T)} \prod_{t=0}^T \sqrt{p_{\theta}^{(t)}(y_t) p_{\theta'}^{(t)}(y_t)} = \prod_{t=0}^T \sum_{y_t \in \mathcal{Y}_t} \sqrt{p_{\theta}^{(t)}(y_t) p_{\theta'}^{(t)}(y_t)}.$$

Singularity. The proof uses the standard inequality $\|p - q\|_{\text{TV}} \leq \sqrt{2} H(p, q)$ on finite spaces, with $H^2(p, q) = 1 - \rho(p, q)$. If $\|p - q\|_{\text{TV}} \geq \underline{\varepsilon}$, then

$$\rho(p, q) \leq 1 - \frac{\underline{\varepsilon}^2}{2}.$$

At each $t \in \mathcal{S}_{\theta, \theta'}$, $\rho_t \leq 1 - \underline{\varepsilon}^2/2 < 1$. As $\mathcal{S}$ is infinite, the product contains infinitely many factors uniformly less than 1: $\prod_{t \in \mathcal{S}} \rho_t = 0$. By the criterion of ?, the infinite product measures are mutually singular (?). *Q.E.D.*

THEOREM S8.2—Posterior consistency under product-measure singularity: *Suppose Θ_K finite, prior μ_0 with full support, and PBE-consistent Bayesian updating (Definition ??). Let θ_K^* be the realized type and $\mathcal{T}$ the prolonged captivity event (Definition ??). Suppose that for all $\theta \neq \theta_K^*$ the following hold:*

- (i) **Trajectory compatibility.** $\mathbb{P}_{\mathcal{C}, \theta_K^*}(\mathcal{T}) > 0$; for each false type, either $\mathbb{P}_{\mathcal{C}, \theta}(\mathcal{T}) = 0$, or signal laws are compared under the law restricted to $\mathcal{T}$.
- (ii) **Product structure and recurrent TV separation.** The sequence $\{Y_t\}_{t \geq 0}$ satisfies product factorization and the existence of $\mathcal{S}_{\theta, \theta_K^*}$ and $\underline{\varepsilon}_{\theta} > 0$ from Lemma ??.
- (iii) **Signal-experiment alignment.** In $\mathcal{T}$, ζ_t contains $Y_t = (m_t, d_t)$, so that singularity of $\{Y_t\}$ transfers to $\{\zeta_t\}$ by monotonicity of singularity with respect to σ -algebras.

Then, in $\mathcal{T}$,

$$\mu_t(\theta) \xrightarrow{t \rightarrow \infty} 0 \quad \forall \theta \neq \theta_K^*, \quad \mu_t(\theta_K^*) \xrightarrow{t \rightarrow \infty} 1, \quad \mathbb{P}_{\mathcal{C}, \theta_K^*}\text{-a.s.}$$

Equivalently, $\{\mu_t\}$ converges weakly to $\delta_{\theta_K^*}$ $\mathbb{P}_{\mathcal{C}, \theta_K^*}$ -a.s. in $\mathcal{T}$.

PROOF: Fix $\theta \neq \theta_K^*$.

Case $\mathbb{P}_{\mathcal{C}, \theta}(\mathcal{T}) = 0$: the false type cannot generate trajectories in $\mathcal{T}$; its posterior likelihood is annulled by survival exclusion.

Case $\mathbb{P}_{\mathcal{C}, \theta}(\mathcal{T}) > 0$: by (ii), Lemma ?? implies that the infinite product laws are mutually singular. By (iii), that singularity transfers to the public flow $\{\zeta_t\}$. There exists a tail event A_{θ} with

$$\mathbb{P}_{\mathcal{C}, \theta_K^*}(A_{\theta} \mid \mathcal{T}) = 1, \quad \mathbb{P}_{\mathcal{C}, \theta}(A_{\theta} \mid \mathcal{T}) = 0.$$

The likelihood ratio R_t^θ of the false type against the true type satisfies $R_t^\theta \rightarrow 0$ $\mathbb{P}_{C, \theta_K^*}$ -a.s. in $\mathcal{T}$ (??). By Bayes,

$$\mu_t(\theta) \leq \frac{\mu_0(\theta)}{\mu_0(\theta_K^*)} R_t^\theta \rightarrow 0 \quad \mathbb{P}_{C, \theta_K^*}\text{-a.s. in } \mathcal{T}.$$

As Θ_K is finite, all false coordinates converge to zero simultaneously. Finally,

$$1 = \lim_{t \rightarrow \infty} \sum_{\theta} \mu_t(\theta) = \lim_{t \rightarrow \infty} \mu_t(\theta_K^*) + 0,$$

so that $\mu_t(\theta_K^*) \rightarrow 1$ $\mathbb{P}_{C, \theta_K^*}$ -a.s. in $\mathcal{T}$.

Q.E.D.

S9. FORMAL VERIFICATION IN LEAN 4

This replication package includes a standalone Lean 4 project in folder `lean/` (main file `Appendix3Proofs.lean`) that formalizes the logical dependency graph of this supplement. The development is independent of earlier project-specific files and mirrors Sections ??–??. Each declaration is tagged with its TeX label where applicable. A successful build (`lakebuild`) yields no `sorry` obligations. Build instructions appear in Section ??.

Reading the status tags. Every Lean declaration carries one of two tags. `[Proved]` means the statement is fully verified by the Lean 4 compiler against Mathlib: the proof is machine-checked and admits no gaps. `[Hypothesis]` means the statement is declared as an explicit interface and *assumed*, not re-proved, inside Lean; each such interface corresponds to a classical step proved analytically in this supplement or in the cited literature. The formalization therefore does *not* replace the proofs in this document: it certifies the finite-dimensional algebra and the logical assembly of the identification argument, and makes every classical input explicit.

Proved in Lean (machine-checked).

- **Section ??.** Lemma ?? and Corollary ??.
- **Section ??.** Public filtration, predictive likelihood, and martingale beliefs; Lemma ?? (bounded martingale with a.s. limit in $[0, 1]$, via Doob’s theorem in Mathlib); and the normalization closure of step 3 of Theorem ??: false-type collapse forces $\mu_t(\theta^*) \rightarrow 1$.
- **Section ??.** Continuation probability of Definition ??; Cox hazard-ratio algebra (distinct $\beta_{K,j} \Rightarrow \text{ratio} \neq 1$); the predictive masses $p_\theta^{(t)}$ form probability masses; local TV separation — Lemma ??(i) — for those masses under $\beta_{K,j}$ separation and a non-knife-edge total hazard; logit axioms (A1) full support and (A2) plan dominance of Remark ??.
- **Section ??.** Proposition ??: compact and signed forms of κ_h (`kappa_h_signed`, `dLambda_eq_kappa`); continuation factorization and (??) (`dContProb_eq`); sign certificate of (??) with explicit Δt and survival tail (`prop_expected_captivity_gamma`, `prop_expected_captivity_sgn`). Wald identity and DCT remain analytic.
- **Section ??.** Logical assembly of Lemma ??: conditions (a)/(b) imply Definition ??; packaging of recurrent uniform TV margins; and the Bayes factorization of step 2 of Theorem ??: the bound $\mu_t(\theta) \leq (\mu_0(\theta)/\mu_0(\theta^*)) R_t^\theta$ forces $\mu_t(\theta) \rightarrow 0$ whenever $R_t^\theta \rightarrow 0$.
- **Section ??.** Bhattacharyya product identity (??); the Hellinger–TV bound $\rho \leq 1 - \varepsilon^2/2$; and vanishing of the product affinity under recurrent separation, i.e., the entire algebraic part of Lemma ??.
- **Identification bridge.** Cox predictive masses plus a recurrent TV margin imply asymptotic identification, conditional only on the two Kakutani interfaces listed below.

Not proved in Lean (explicit hypotheses). The following classical steps are stated as interfaces and assumed inside Lean, mainly because infinite product measures and their convergence theory are not available in Mathlib. Each one is proved analytically where indicated.

- *Kakutani mutual singularity* — vanishing product affinity implies mutually singular trajectory laws: proof of Lemma ?? and ?.
- *Singularity transfer* $Y_t \Rightarrow \zeta_t$: item (iii) of Theorem ??.
- *Singularity implies vanishing likelihood ratio*: step 2 of Theorem ?? (?).
- *Survival exclusion*, $\mathbb{P}_{C,\theta}(\mathcal{T}) = 0$ implies collapse: step 2 of Theorem ??.
- *Maximum entropy axiom* (A3) of Remark ??: ?.
- *Implicit-function step* of Proposition ??: proof in Section ??.
- *Existence of the implementable PBE*, Theorem ??: fixed-point proof in Section ?? (?).

Table ?? maps these hypotheses to their Lean identifiers.

TABLE S9.1
LEAN FORMALIZATION STATUS FOR SELECTED SUPPLEMENT OBJECTS

TeX object	Lean identifier	Status
Lemma ?? (algebraic part)	generalIdAppendixLemma	Proved
Kakutani (1948) mutual singularity	kakutaniMutualSingularity	Hyp.
Outcome-trajectory laws on $\mathcal{T}$	kakutaniSignalSingularity	Hyp.
Singularity transfer $Y_t \Rightarrow \zeta_t$	signalExperimentAlignment	Hyp.
Singularity $\Rightarrow$ vanishing likelihood ratio	singularityImpliesLikelihoodVanishing	Hyp.
$\mathbb{P}_{C,\theta}(\mathcal{T}) = 0 \Rightarrow$ collapse	zeroMassPosteriorCollapse	Hyp.
Remark ??, axiom (A3)	logitProb_maxEntropy_hyp	Hyp.
Proposition ??, eq. (??) (signed form)	kappa_h_signed, dLambda_eq_kappa	Proved
Proposition ??, eq. (??)	dContProb_eq	Proved
Proposition ??, eq. (??) (sign)	prop_expected_captivity_gamma, prop_expected_captivity_sgn	Part.
Proposition ??	prop_optimal_gamma_beliefs	Hyp.
Theorem ??	conditionalImplementability	Hyp.

Note. [Proved]: machine-checked in Lean 4. [Part.]: partial certificate (sign of (??) given the analytic chain). [Hyp.]: declared interface; proved analytically in this supplement or cited literature. *Source:* Analytical proofs in this supplement; Lean identifiers in `lean/Appendix3Proofs.lean`. Software: Lean 4 (lean-toolchain); Mathlib via Lake.

S10. REPRODUCIBILITY

All analytical proofs in this supplement are self-contained in `tex/Appendix_3.tex`. Machine-checked algebraic steps are reproduced by the Lean 4 project in `lean/`. Set the working directory to `Appendix_3/` and run `replicate.sh`. See `README.txt` at the package root for step-by-step instructions. Requirements: Lean 4 (toolchain in `lean-toolchain`) and Lake; Mathlib is fetched automatically on first build (network required; estimated first-build time 15–45 minutes).

1. `replicate.sh` at the package root runs `lakebuild` in `lean/` and writes a log to `lean/build.log`.
2. Main formal file: `lean/Appendix3Proofs.lean`. Project metadata: `lakefile.lean`, `lean-toolchain`, `lake-manifest.json`.
3. Section ?? and Table ?? document which results are [Proved] vs. [Hyp.].
4. (Optional) Compile `tex/Appendix_3.pdf` from `tex/` with `pdflatex` and `bibtex`.

Section ?? (Table ??) documents which results are machine-checked vs. hypothetical. All paths are relative to `Appendix_3/`; see `README.txt` for the full build workflow.